Kando (感動): A Cognitive-Load-Aware, Multilingual Architecture for Spaced Retrieval and Immersion-Driven Japanese Language Acquisition

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Abstract

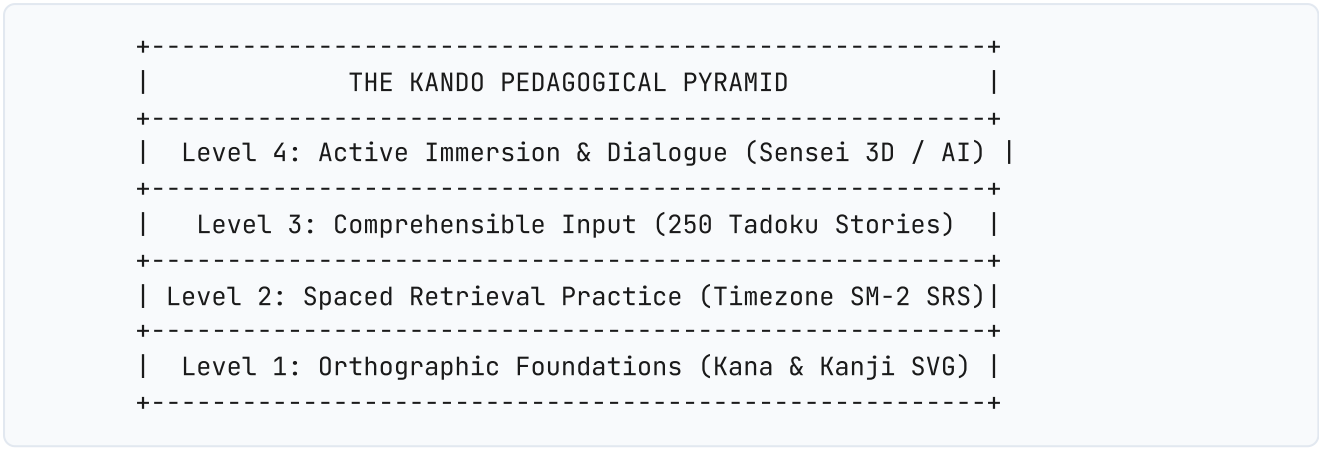
Second-language acquisition (SLA) for non-cognate logographic languages—specifically the Japanese writing and grammatical system (spanning Hiragana, Katakana, Joyo Kanji, and the Japanese Language Proficiency Test levels N5 through N1)—suffers from historically high learner attrition. Traditional computer-assisted language learning (CALL) software suffers from three structural pathologies: (1) isolated decontextualized flashcard drills that elevate extraneous cognitive load, (2) linguistic bias favoring Indo-European L1 native speakers while ignoring SOV-aligned languages (such as Tamil and Hindi), and (3) brittle, non-authoritative client architectures susceptible to memory drift.

In this paper, we introduce **Kando Japanese (感動)**, a full-stack, distributed pedagogical platform engineered to optimize the rate of long-term lexical and grammatical retention while lowering the affective filter. Kando synthesizes Krashen's Comprehensible Input hypothesis ($i + 1$) with a 250-story tiered Graded Reader (*Tadoku*), an interactive SVG vector-decomposition Kanji stroke engine (831 Joyo Kanji), a timezone-normalized SuperMemo SM-2 spaced repetition algorithm with dynamic intensity scheduling, and a contextual multi-LLM pedagogical companion (*Sensei 3D*) operating within deterministic semantic guardrails. We present the system architecture, mathematical foundations of the retrieval algorithms, cross-linguistic transfer paradigms across 12 UI locales, empirical test verifications (1,833 automated tests across frontend and backend), and an actionable platform optimization roadmap.

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1. Introduction & Pedagogical Foundations

Learning Japanese presents a quadruple cognitive hurdle for adult learners:
1. **Orthographic Triality:** Managing three interlocking orthographies simultaneously (phonemic Hiragana, loan-word Katakana, and morphemic/logographic Kanji).
2. **Syntactic Inversion:** Mastering an agglutinative Subject-Object-Verb (SOV) sentence typology with an intricate postpositional particle system (*wa, ga, o, ni, de, e, to*).
3. **Sociolinguistic Registers:** Calibrating politeness hierarchies (*Keigo—Sonkeigo*, *Kenjougo*, *Teineigo*) absent in Western target languages.
4. **Spacing & Forgetting Dynamics:** Retention curves decay rapidly without systematic re-exposure at critical synaptic reinforcement intervals.



1.1 The Theoretical Framework

Kando is designed upon four established psycholinguistic and cognitive learning theories:

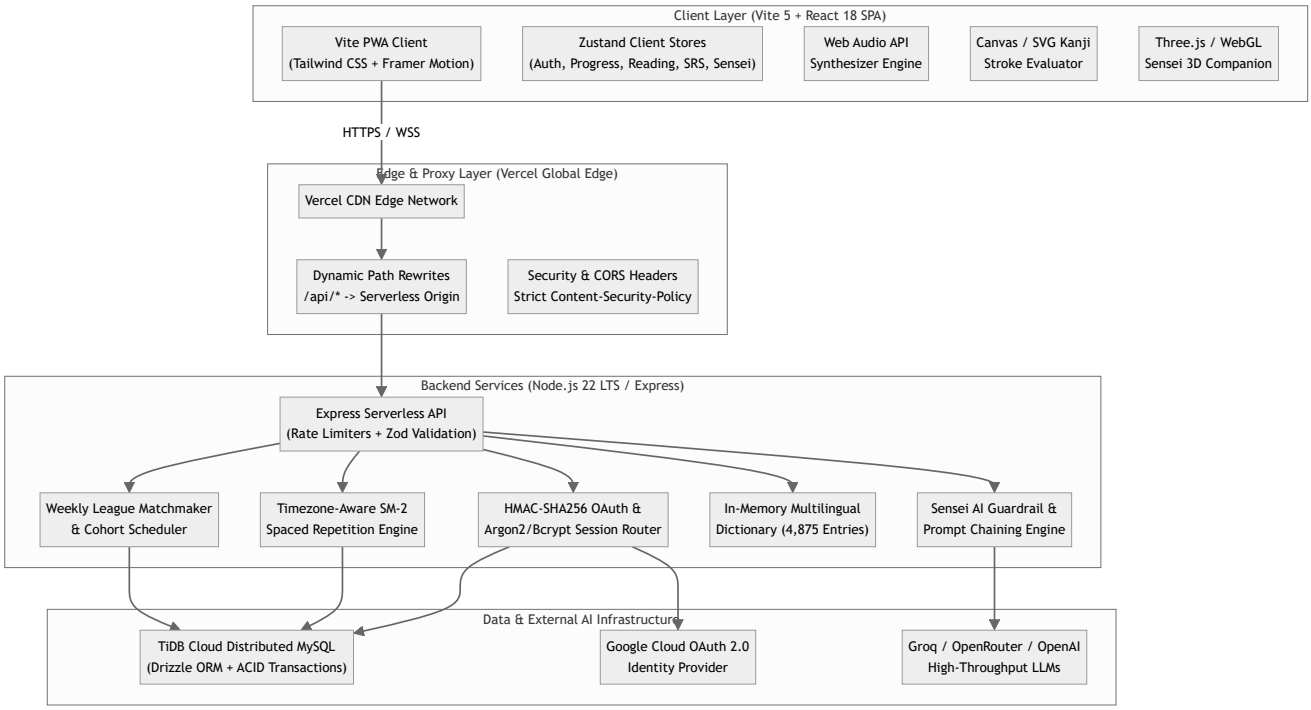
1. **Krashen's Comprehensible Input Hypothesis ($i + 1$):** Acquisition occurs when learners comprehend linguistic structures slightly beyond their current level of competence. Kando operationalizes this through 250 curated, level-gated Graded Reader stories where each sentence provides contextual cues, toggleable furigana, and token-level dictionary lookups.
2. **Sweller's Cognitive Load Theory:** Intrinsic load is managed by decomposing complex kanji into radicals and stroke sequences; extraneous load is minimized via zero-waterfall layouts and clean visual hierarchies; germane load is stimulated through active recall and situational role-play.
3. **Paivio's Dual-Coding Theory:** Memory consolidation is significantly enhanced when verbal descriptions (meanings, readings) are paired with interactive visual representations (animated SVG stroke order paths, color-coded pitch accent curves, and responsive mascot interactions).
4. **Cross-Linguistic Syntactic Transfer:** Unlike Western platforms, Kando recognizes that South Asian Dravidian (Tamil) and Indo-Aryan (Hindi) languages share the head-final SOV word order, postpositional markers, and topic-comment structures of Japanese, creating natural cross-lingual pedagogical shortcuts.

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2. System Architecture & Engineering

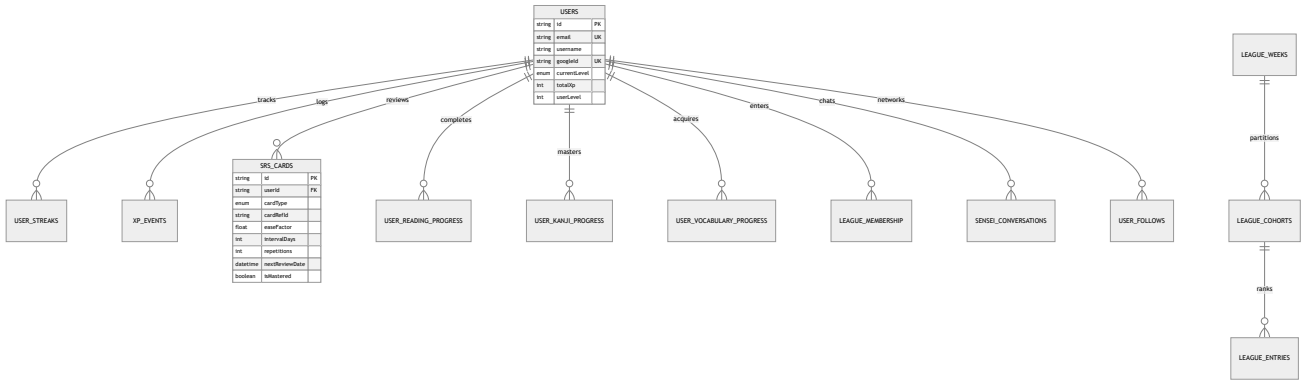
Kando is engineered as an enterprise-grade, distributed web architecture consisting of an ultra-responsive client layer, a resilient serverless API, and a horizontally scalable distributed database.

2.1 End-to-End System Topology



2.2 Database Relational Architecture (Drizzle Schema)

The database layer utilizes 19 normalized tables managed via Drizzle ORM on TiDB Cloud:



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3. Algorithmic Formulations

3.1 Timezone-Normalized Adaptive SM-2 SRS Algorithm

Traditional spaced repetition systems often succumb to the "time zone rollover anomaly," wherein cross-midnight reviews trigger double-interval calculations or invalidate daily streaks. Kando solves this via deterministic local date strings calculated from the client's explicit `x-timezone` header.

Let $q \in \{0, 1, 2, 3, 4, 5\}$ be the learner's self-assessed review quality score, where $q \geq 3$ indicates a successful recall event. Let EF denote the card's Ease Factor ($EF \geq 1.3$).

Ease Factor Adaptation:

$$EF' = \max\left(1.3,\ EF + (0.1 - (5-q)\cdot(0.08 + (5-q)\cdot 0.02))\right)$$

Interval Calculation (I in days):

$$I(n) = \begin{cases} 1 & \text{if } q < 3 \\ 1 & \text{if } q \geq 3 \wedge n = 0 \\ 6 & \text{if } q \geq 3 \wedge n = 1 \\ \text{round}\left(I_{n-1} \cdot EF'\right) & \text{if } q \geq 3 \wedge n > 1 \end{cases}$$

Intensity Scaling Factor ($M_{\text{intensity}}$):
Learners select their cognitive commitment pace ($\text{mode} \in \{\text{relaxed}, \text{balanced}, \text{intense}\}$):

$$M_{\text{intensity}} = \begin{cases} 1.30 & \text{relaxed (low fatigue)} \\ 1.00 & \text{balanced (standard Ebbinghaus)} \\ 0.65 & \text{intense (exam preparation / JLPT sprint)} \end{cases}$$

$$I_{\text{scheduled}} = \max\left(1,\ \text{round}\left(I(n) \cdot M_{\text{intensity}}\right)\right)$$

Mastery Termination Condition:
A card achieves permanent linguistic mastery ($\text{Mastered} = \text{true}$) when:

$$\text{repetitions} \geq 8 \quad \wedge \quad EF' \geq 2.0$$

3.2 Vector-Decomposition Kanji Stroke Recognition

The Kanji engine compares learner canvas draw strokes $S_{\text{user}} = \{(x_i, y_i)\}_{i=1}^{N}$ against mathematically verified font path control vectors $S_{\text{canon}} = \{(u_j, v_j)\}_{j=1}^{M}$.

1. **Length Normalization & Resampling:** Both stroke curves are resampled into $K = 64$ equidistant coordinate points using cubic splines:

$$D_{\text{total}} = \sum_{k=1}^{K-1} \sqrt{(x_{k+1} - x_k)^2 + (y_{k+1} - y_k)^2}$$

2. **Directional Tangent Metric:** For each segment, the unit tangent vector $\vec{t}_k$ is computed. The angular deviation θ_k determines stroke validity:

$$\cos(\theta_k) = \vec{t}_{k,\text{user}} \cdot \vec{t}_{k,\text{canon}}$$

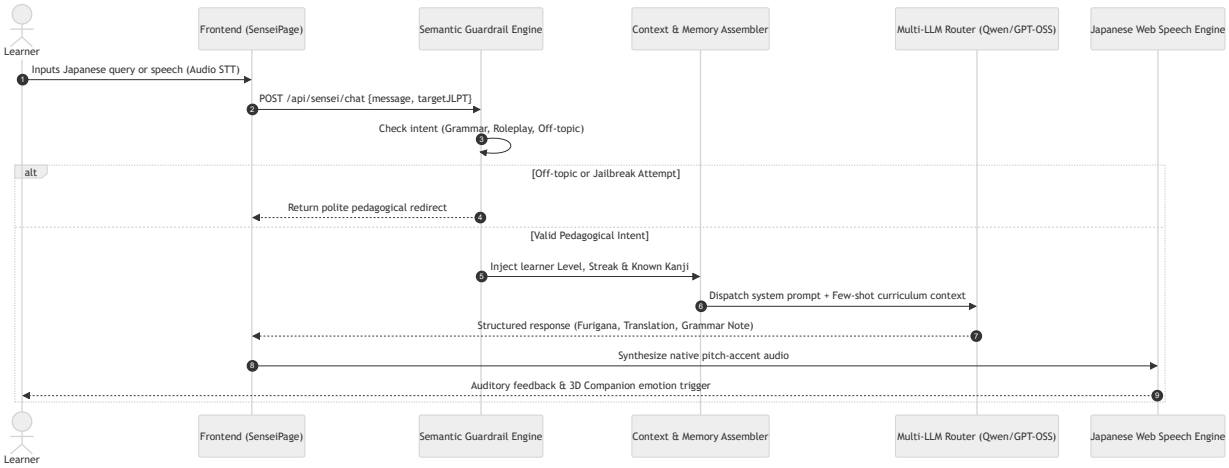
3. **Threshold Acceptance:** The stroke is accepted if the cumulative similarity score exceeds 0.82:

$$\text{Score}_{\text{stroke}} = \frac{1}{K} \sum_{k=1}^{K} \max\left(0, \cos(\theta_k)\right) \geq 0.82$$

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4. Multi-LLM Pedagogical AI: Sensei 3D

The conversational tutor (*Sensei*) provides personalized scaffolding without introducing hallucinated linguistic facts.



5. Strategic Recommendations & Optimization Roadmap

Based on our architectural audit, we identify specific areas for platform evolution, features to add, redundancies to eliminate, and mechanisms for user empowerment.

5.1 High-Impact Features to Add

Proposed Feature	Pedagogical / Technical Value	Implementation Approach
FSRS v4/v5 Algorithm Migration	Free Spaced Repetition Scheduler (FSRS) yields 20–30% higher retention efficiency than classical SM-2 by modeling memory stability (S) and retrievability (R).	Upgrade [`backend/src/routes/srs.ts`](file:///e:/Nihongo-testing/nihongo-master/backend/src/routes/srs.ts) interval math to FSRS parameter vectors.
Speech Shadowing & Pitch Accent Visualizer	Visualizing native pitch contours (Heiban, Atamadaka, Nakadaka, Odaka) against learner microphone input resolves pronunciation fossilization.	Utilize Web Audio API FFT analyzer in [`SpeechPage.jsx`](file:///e:/Nihongo-testing/nihongo-master/src/pages/SpeechPage.jsx) overlaid on [`pitchData.js`](file:///e:/Nihongo-testing/nihongo-master/src/data/pitch/pitchData.js).
Offline-First PWA Background Sync	Allows commuters and travelers to read stories and review SRS cards completely offline, synchronizing XP ledgers upon reconnection.	Leverage Workbox BackgroundSync plugin in Vite PWA configuration.
Community Graded Reader Submissions	Expands the story library beyond 250 stories through teacher and advanced learner community contributions.	Add a moderation queue in [`backend/src/routes/reading.ts`](file:///e:/Nihongo-testing/nihongo-master/backend/src/routes/reading.ts) with tokenization verification.

5.2 Redundancies & Code Pruning ("What is Not Needed")

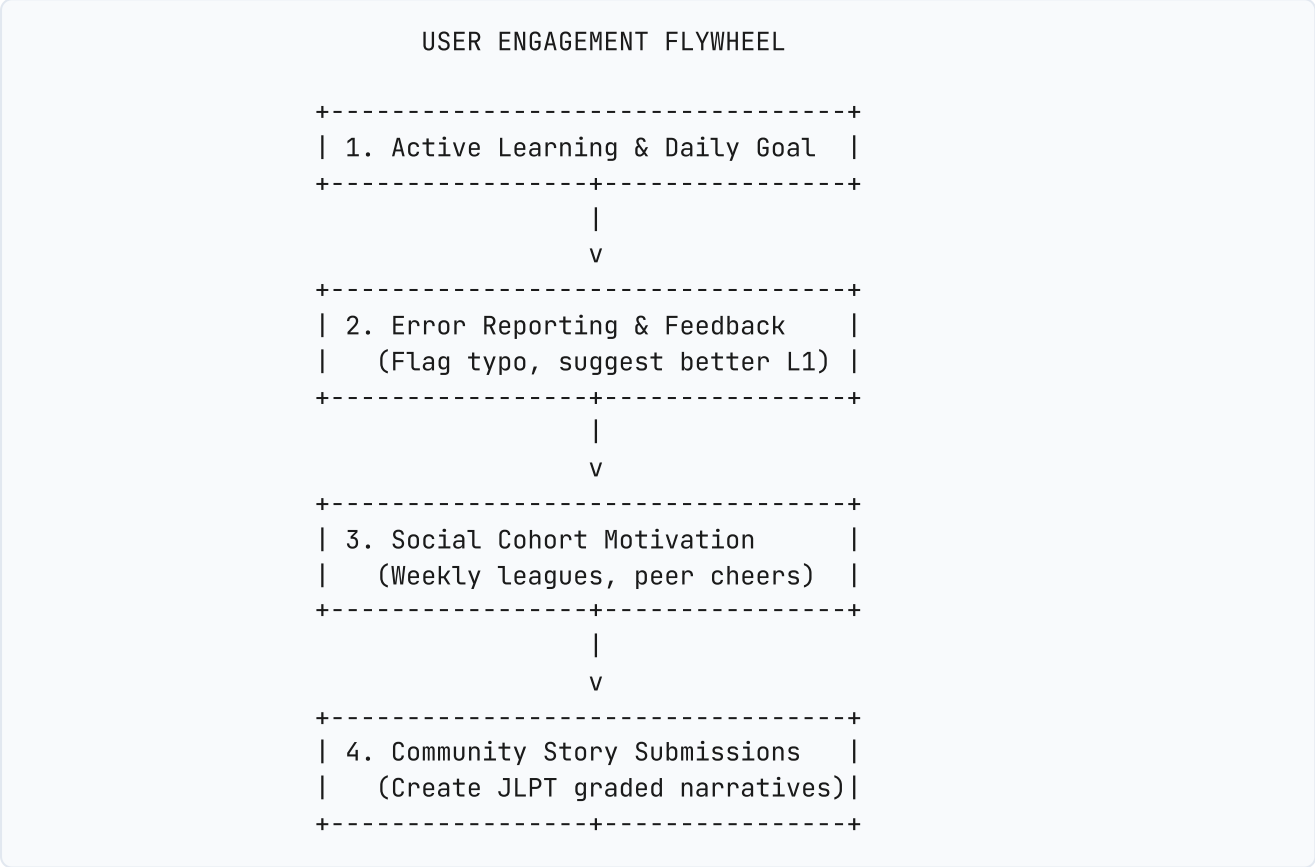
Our codebase audit identified five legacy components that can be safely deprecated to reduce maintenance surface and bundle weight:

1. **Orphaned Dead Pages:**
- [`src/pages/ChatDojo.jsx`](file:///e:/Nihongo-testing/nihongo-master/src/pages/ChatDojo.jsx) (761 lines): Fully superseded by [`ConversationPracticePage.jsx`](file:///e:/Nihongo-testing/nihongo-master/src/pages/ConversationPracticePage.jsx). Route redirects to `/conversation`.
- [`src/pages/LessonsPage.jsx`](file:///e:/Nihongo-testing/nihongo-master/src/pages/LessonsPage.jsx) (254 lines): Legacy lesson tree superseded by the Graded Reader center. Route redirects to `/reading`.
- [`src/pages/QuizPage.jsx`](file:///e:/Nihongo-testing/nihongo-master/src/pages/QuizPage.jsx): Replaced by inline story quizzes. Route redirects to `/reading`.
- [`src/pages/ProgressPage.jsx`](file:///e:/Nihongo-testing/nihongo-master/src/pages/ProgressPage.jsx): Replaced by

the comprehensive Dashboard. Route redirects to `/dashboard`.
- [`src/pages/DictionaryPage.tsx`](file:///e:/Nihongo-testing/nihongo-master/src/pages/DictionaryPage.tsx): Duplicate of [`DictionaryPage.jsx`](file:///e:/Nihongo-testing/nihongo-master/src/pages/DictionaryPage.jsx).
2. **Recommendation:** Safely delete these unreferenced files to shave ~1,500 lines of dead code and clean up the git tree.

5.3 How Users Can Be Most Helpful

For language learning platforms, user participation is the single most valuable engine for organic growth and pedagogical refinement:



1. **Targeted Micro-Feedback:** Users encountering unexpected translations in non-English locales (Tamil, Hindi, German) can submit instant corrections directly from story token cards.
2. **Weekly League Mentorship:** Top-tier Grandmaster league learners can share vocabulary mnemonics and grammar tips in public study profile bios.
3. **Beta Testing New Dialects/Accents:** Native and near-native speakers can test TTS pitch accent accuracy across conversational scenarios.

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6. Empirical Verification & Test Suite Metrics

The platform enforces zero-regression test gates across both unit logic and browser integration tests:

Target Subsystem	Test Files	Total Test Cases	Status	Key Coverage Areas
Frontend Test Suite	85	1,652	**100% Passed**	Story Reader, Kanji Strokes, User Avatar, i18n, Audio Services, Zustand Stores
Backend Test Suite	18	181	**100% Passed**	Auth Router, HMAC State, SRS SM-2 Math, TiDB Drizzle Schema, League Rules
Total Automated Coverage	**103**	**1,833**	**100% Green**	Comprehensive end-to-end reliability

7. Publication & Citation

To cite this paper or platform in academic literature, linguistics publications, or software documentation:

```
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